

ALCHEMY AOE ALLIANCE

**TYPE:
HANDBOOK
(HDBK)**



**PROJECT:
N/A**

HDBK--L008

REVISION: R3

Handbook, Alchemy League Season 8

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1 - Referenced Documents

Table 1 below is a list of all specifications that are components to this one. Each of its entries were designed as an interchangeable part, to be used in context for this specification, or others like it:

Specification Number	Nomenclature
<u>MTHD-002</u>	Calculation, Tournament Elo
<u>MTHD-003</u>	Elo Correction, Handicap
<u>MTHD-004</u>	Discipline, Pa2 Round Robin
<u>MTHD-005</u>	Winner Selection, Round Robin
<u>MTHD-006</u>	Earnings, Round Robin
<u>MTHD-007</u>	Structure, Single Division Stack League
<u>STND-005</u>	Competition, General Rules

Table 1: Bill of Materials

If any part of any documents listed in Table 1 conflict with this specification, then this specification shall assume priority.

2 - Background

The *Alchemy Alliance (ALAL)* is a free and non-profit organization of random map scripters, tournament hosts, content creators, and skilled players dedicated to the advancement of custom random maps in Age of Empires II: Definitive Edition. As the nucleus of the *Alchemy AOE Community*, the *ALAL* publishes quality standards for custom random maps, and curates the finished products to be featured into thematic competitions where adventuresome players may exhibit creative strategies for maximum entertainment value.



3 - Overview

This handbook describes Alchemy League, an Age of Empires II 1v1 competition hosted by the *Alchemy Alliance* with the following unique features, summarized in reference below:

- Participants will generally not face an opponent 300 elo higher than they are without handicap.
- Cutting-edge tournament elo calculation from online match history, based on weighted average and refactored to account for win rate. Player statistics also adjusted according to performance in previous league seasons, where required.
- Handicap used sparingly to include extremities of bell curve, so that no player will be turned away based on skill.
- Unique bracketing structure & prize distribution.
- Standard Victory Condition on a Custom Map Pool.

4 - Prize Money

Prize Money Details per Table 4 below

Title	Amount	Details
Winners per Sprint	8	Top Performers per <u>MTHD-005</u>
Prizes, Players of Sprint 1	\$92 USD	Distribution per <u>MTHD-006</u>
Prizes, Players of Sprint 2	\$172 USD	
Skill Bias Exponent	1.25	
Payment Platform	~	<u>Paypal</u> (direct USD) or Solana (indirect, but cheaper) transfer to platform of choice (Paypal, Kraken, Coinbase, Binance, etc.)
Country of Payment Origin	~	United States of America

Table 4: Prize Money Information



5 - Timeline

Start Date (YY-MM-DD)	End Date (YY-MM-DD)	Milestone(s):
~	2025-09-11	Community Classic Map Adjustments
~	2025-09-14	Registrations Close
2025-09-17	2025-09-30	Round 1 (Sprint 1)
2025-09-24	2025-10-07	Round 2 (Sprint 1)
2025-10-01	2025-10-14	Round 3 (Sprint 1)
2025-10-08	2025-10-21	Round 4 (Sprint 1)
2025-10-22	2025-10-28	Break, Calculate Winnings
2025-10-29	2025-11-11	Round 5 (Sprint 2)
2025-11-05	2025-11-18	Round 6 (Sprint 2)
2025-11-12	2025-11-25	Round 7 (Sprint 2)
2025-11-19	2025-12-02	Round 8 (Sprint 2)
2025-12-03	2025-12-09	End, Calculate Winnings

Table 5: Timeline, Alchemy League, This Season



6 - League Format

Title	Min	Max	Details
Team Size	~	~	1v1
Number of Divisions	1	1	Round Robin per <u>MTHD-007</u> .
Number of Sprints	2	2	
Rounds per Sprint	4	4	
Games per Round	2	3	After completing the minimum number of required games, players may mutually agree to continue playing until the maximum number of games has been reached, regardless of results from previous games. Submission of a recorded game file is the only accepted evidence of consent. All games played in this format will be used to calculate league results and payout.

Table 6: League Format Information



7 - Tournament Elo

Title	Details
Data Collection, Step 1	Reference internet APIs to garner all alternate or “smurf” accounts corresponding to the same player, derived through IP address, Steam account sharing, and other methods.
Data Collection, Step 2	Reference internet APIs to harvest all relevant match history corresponding to each of the accounts gathered from the previous step.
Average Ranked Elo, Formula	Calculated from 100 most recent games, linearly weighted per <u>MTHD-002</u> .
Average Ranked Elo, Confidence Window	Perform a tolerance sweep on win rate and elo fluctuation over history to estimate a range of possible tournament elos for each participant, with 100% confidence interval.
Average Ranked Elo, Final Value	Take average ranked elo as the $\frac{3}{4}$ point going from minimum to maximum within confidence window.
Tournament Elo, final Value	Average Ranked Elo ($\frac{3}{4}$ point) adjusted for returning players inactive in ranked.
Handicap	For each matched set of players, 100 shall be added to the lower of two elos and then <u>MTHD-003</u> applied to determine corrective handicap.

Table 7: Tournament Elo Information



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8 - Custom Map Information			
Title	Details		
Map Qualities	Custom Map titles and properties per Appendix A .		
Featured Map	For each non-wild round, the tournament host will declare a map from the pool to be “Map of the Round”. A map cannot be “Map of the Round” twice in the same season of the league.		
Obtaining Maps	To protect creator intellectual property rights, the Alchemy AOE Community does not upload competition resources to the in-game mod center. Participants shall download the latest Alchemy League map scripts and place them in a folder suitable for integration into the game. Recommended approach (~30 seconds): download and run the following open-source batch script: Download .bat file here. This executable was written by the community to automatically download and move the latest version of each map in the pack from the online repository to the correct location. It is used by most players. Direct control (~2 minutes): download the “Map_Archive” zip file from the Alchemy AOE Season web-page and unzip to the appropriate game folder. Note that the direct download is required for non-Windows users. The Alchemy AOE Community server is webhook-configured to notify players of any changes to the random map Github repository –the source of the custom map pool. Players are responsible for making sure they are using the latest versions by re-downloading after a map is updated. Tournament staff are eager to assist as needed.		
	Map Selection, Round Wild	For the Round Wild, both players declare a home map. Then, they take turns discarding the remaining maps. If a third game is played, it shall be on the map discarded last.	
	Map Selection, Rounds Normal	The first map is the “Map of the Round”. Then, players take turns discarding the remaining maps until only two remain. The second game is played on a map randomly selected from the remainder. The unchosen map shall be used for a third game, if played. Draft Link provided in Tournament Server.	
<i>Table 8: Custom Map Information</i>			
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9 - General Settings

Title	Details
Civilization Selection	Civilization Draft consists of five administrative bans, followed by 2 bans each, then 4 picks each, then one snipe each (hidden and parallel), and ends with a pick each.
Tournament Server	https://discord.gg/h9DvzHBNPr
Tournament Host	TechChariot
General Rules	Per <u>STND-005</u>, with the following exceptions: Table 9: • Game Mode → King of the Hill (instead of “Random Map”) • Sudden Death Mode? → Yes (instead of “No”) • Regicide Mode? → Yes (instead of “No”) Figure 9 does not portray these exceptions, and serves no purpose beyond reminding players to check the “Antiquity” mode box.
Discipline	Administrative penalties assigned per <u>MTHD-004</u> .
First Sprint Eligibility	Participants shall have 100 1v1 games minimum and win rate between 15% and 85% totaled across all accounts.
Other Sprint Eligibility	Participants shall complete all previous Sprints without accumulation of excessive administrative penalties.
First Sprint Registration, Access Link	https://forms.gle/ZBxGQufXkSEELcSZ6
First Sprint All Signups, Access Link	https://docs.google.com/spreadsheets/d/1HfdG7YO-ql1TjHx37bGE6BMf6OTyuVJITcMh9l7DlpQ/edit?resourcekey=&gid=993490506#gid=993490506
Other Sprint Registration	Eligible players auto-enrolled in subsequent Sprints unless otherwise indicated at least 24 hours before start.

Table 9: General Settings



Signatures:

TechChariot

2025-03-29

Prepared

*Date (YYYY-MM-DD)*CookiePirate

2025-07-12

Checked

*Date (YYYY-MM-DD)*Honeybadger

2025-07-12

Approved

Date (YYYY-MM-DD)

Revision History:

<u>Revision</u>	Description	Change Document	Date (YYYY-MM-DD)																													
X1	-- Advanced Release for Community Review --	N/A	2025-05-15																													
X2	Added complete and accurate timetable, including map development period.	N/A	2025-05-19																													
OI	---- Original Issue ----	N/A	2025-07-12																													
R1	§3 “Overview”, Bullet 2: ADDED second sentence. Table 4: Removed Row 5: <table><tr><td>Prizes, Map Creators</td><td>\$120 USD</td><td>Distribution per <u>PSRM-PC-1</u></td></tr></table> Table 5: Removed first eight rows: <table><tr><td>~</td><td>2025-07-13</td><td>Specification Package Released</td></tr><tr><td>2025-07-14</td><td>2025-07-21</td><td>Map Nominations Open</td></tr><tr><td>2025-07-22</td><td>2025-07-25</td><td>Map Nomination Check Phase</td></tr><tr><td>2025-07-26</td><td>2025-08-02</td><td>Map Feedback Phase</td></tr><tr><td>2025-08-03</td><td>2025-08-17</td><td>Map Update Phase</td></tr><tr><td>2025-08-18</td><td>2025-08-22</td><td>Map Inspection Phase</td></tr><tr><td>2025-08-23</td><td>2025-08-25</td><td>Map Slideshow Preparation</td></tr><tr><td>2025-08-26</td><td>2025-09-09</td><td>Map Voting Phase</td></tr></table> Table 5: Revised Column 3, Row 9 FROM: Release PSRM-003 TO: Community Classic Map Adjustments Table 7: Rows 3-5, Column 1 and Row 5, Column 2: Revised FROM “...Tournament...” TO “Average Ranked”. Table 7: ADDED Row between Row 5 and 6.			Prizes, Map Creators	\$120 USD	Distribution per <u>PSRM-PC-1</u>	~	2025-07-13	Specification Package Released	2025-07-14	2025-07-21	Map Nominations Open	2025-07-22	2025-07-25	Map Nomination Check Phase	2025-07-26	2025-08-02	Map Feedback Phase	2025-08-03	2025-08-17	Map Update Phase	2025-08-18	2025-08-22	Map Inspection Phase	2025-08-23	2025-08-25	Map Slideshow Preparation	2025-08-26	2025-09-09	Map Voting Phase	N/A	2025-08-28
	Prizes, Map Creators	\$120 USD	Distribution per <u>PSRM-PC-1</u>																													
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R1	Table 8: Row 1, Column 2, Revised FROM: ““Player Choice Returning Maps” selected through process of <u>PSRM-002</u>. Resulting map pool defined in <u>PSRM-003</u>.” TO: “Custom Map titles and properties per Appendix A”. Table 9: Row 1, added list of exceptions.	N/A	2025-08-28	
R2	Table 1: Removed final two rows of “Referenced Documents” that should have been removed when R1 was issued. Table 4: Row 2, Column 2: Revised FROM: “80” TO: “92”. Table 4: Row 3, Column 2: Revised FROM: “160” TO: “172”. Table 4: Rows 3, Column 3: Revised FROM: “<u>Paypal</u>” TO: “<u>Paypal</u> (direct USD) or Solana (indirect, but cheaper) transfer to platform of choice (Paypal, Kraken, Coinbase, Binance, etc.)”. APPENDIX A: Revised FROM: “Player-Assigned Fortified Towers have 50% chance of base range 8, and 50% chance of base range 9, different each game.” TO: “Player-Assigned Fortified Towers have 50% chance of base range 9, and 50% chance of base range 10, different each game.” APPENDIX A: Added bullet about starting petard, revised image of start to show petard.	N/A	2025-08-28	
R3	Table 8, Row 3, Column 2: Revised FROM: “Participants shall download and execute the following batch script to automatically gain access to latest Alchemy League maps in game: <u>Download .bat file here</u>.” The Alchemy AOE Community server is webhook-configured to notify players of any changes to the random map Github repository – the source of the custom map pool. Players are responsible for making sure they are using the latest versions by re-running the batch script whenever a relevant map is changed. Note that the batch script works only for Windows users. If a participant has a different operating system, tournament staff will assist as needed.”	N/A	2025-09-04	
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HDBK--L008		Handbook, Alchemy League Season 8		REV: R3	
R3		Table 8, Row 3, Column 2: Revised TO: “Participants shall download the latest Alchemy League map scripts and place them in a folder suitable for integration into the game. Recommended approach (~30 seconds): download and run the following open-source batch script: <u>Download .bat file here</u>. This executable was written by the community to automatically download and move the latest version of each map in the pack from the online repository to the correct location. It is used by most players. Direct control (~2 minutes): download the “Map_Archive” zip file from the Alchemy AOE Season web-page and unzip to the appropriate game folder. Note that the direct download is required for non-Windows users. The Alchemy AOE Community server is webhook-configured to notify players of any changes to the random map Github repository – the source of the custom map pool. Players are responsible for making sure they are using the latest versions by re-downloading after a map is updated. Tournament staff are eager to assist as needed.”	N/A	2025-09-04	

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A. APPENDIX – Map Properties

The following Classic Standard Maps were downloaded from 2020 game data, and prefixed with “MWDMA”, standing for “A Million Ways to Die in the Middle Ages”:

- MWDMA_Arabia
- MWDMA_Archipelago
- MWDMA_Arena
- MWDMA_Black_Forest
- MWDMA_Continental
- MWDMA_Gold_Rush
- MWDMA_Highland
- MWDMA_Mongolia
- MWDMA_Oasis
- MWDMA_Rivers
- MWDMA_Salt_Marsh
- MWDMA_Yucatan

These maps were modified to give players a Fortified Tower at least 30 tiles away from starting Town Center, then rule changes were added to enable the maximum number of possible non-exclusive win conditions, listed below:

- Control Monument Until Years = 0
- Destroy Enemy King
- Destroy Enemy Town Center
- Destroy Enemy Fortified Tower



- Control All Relics Until Years = 0
- Control a Wonder Until Years = 0

The following additional rule changes were added:

- Game aborts after launch if any of the following lobby options were true:
 - Sudden Death box unchecked
 - Regicide box unchecked
 - King of the Hill Game Mode Unselected
 - Empire Wars box checked
 - Resources are not “Standard” or “Low Resources”.

In this way, participants are prevented from accidentally playing on incorrect settings.

- Player-Assigned Fortified Towers have random health between [950, 1425] hit-points, different each game.
- Player-Assigned Fortified Towers have 50% chance of base range 9, and 50% chance of base range 10, different each game.
- Player-Assigned Fortified Towers surrounded by ring of walls of random type (different each game), breakdown as follows:

Probability	Wall Type
35%	Palisade
30%	Fortified Palisade
20%	Stone
15%	Fortified Stone



- Monuments do not provide bonus resource trickle.
- Players start with an unmodified petard instead of a traditional scout.

The following images are provided for reference:



Finally, it should be noted that the maps for this event are procured outside the usual Alchemy AOE Quality system (due to time constraint of revising the ALS8 plan no fewer than thrice), and due to their background as “throwback classics”, they can be expected to result in a number of unfair generations – generally acceptable in 2020 but not by the standards of today.

It is the hope of the Alchemy AOE Alliance that registrants in ALS8 and community members can help identify the most pressing issues in the listed maps, at least eight of which will be selected for play once Season 8 begins. It is also our belief that the numerous win conditions will result in games that end before most unfairness can be exploited.